

AKTOBE, Kazakhstan

WMO: 352290

Lat: 50.2578N

Lon: 57.2236E

Elev: 719

StdP: 14.32

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-15.2	-25.8	1.3	-19.9	-20.7	1.8	-15.3	30.1	20.8	27.2	16.1	3.9	150	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	96.1	65.4	93.0	64.9	89.5	64.0	68.7	87.0	67.3	85.1	66.0	83.4	10.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	88.8	72.6	61.4	83.5	72.4	59.8	78.9	72.1	33.3	87.0	32.1	84.8	31.1	83.3	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	23.2	20.6	DB	-27.6	100.6	7.1	3.1	-32.6	102.9	-36.8	104.7	-40.7	106.5	-45.9	108.8	(4)
(5)			WB	-27.8	71.8	7.0	1.5	-32.8	72.9	-36.9	73.8	-40.9	74.6	-46.0	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	8.1	10.4	23.5	45.7	59.5	69.2	72.8	70.4	57.9	43.4	27.8	14.5		
	DBStd	25.36	12.60	12.69	11.04	10.87	8.10	7.81	6.50	7.81	8.42	8.68	10.82	12.48		
	HDD50	5489	1300	1110	823	207	17	1	0	0	24	241	665	1101		
	HDD65	9054	1765	1530	1287	582	208	44	11	37	239	670	1115	1566		
	CDD50	2602	0	0	1	78	310	578	706	633	261	35	0	0		
	CDD65	694	0	0	0	2	37	171	252	205	27	0	0	0		
	CDH74	9641	0	0	0	75	668	2381	3229	2767	508	13	0	0		
	CDH80	4370	0	0	0	10	204	1108	1518	1350	179	1	0	0		
(14) Wind	WSAvg	7.8	7.7	8.2	9.0	8.8	8.4	7.8	7.2	7.0	6.9	7.4	7.7	8.0		
(15) Precipitation	PrecAvg	12.0	0.9	0.8	0.8	1.1	1.1	1.4	1.2	1.0	0.8	1.1	1.0	1.0		
	PrecMax	16.0	2.2	2.6	1.9	2.4	3.1	3.9	4.3	3.0	3.0	3.2	2.4	2.6		
	PrecMin	7.6	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.1	0.0		
	PrecStd	2.4	0.5	0.6	0.6	0.7	0.8	0.9	0.9	0.8	0.6	0.7	0.5	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	35.6	58.6	81.5	89.4	98.2	99.0	100.4	93.6	75.2	53.6	37.6		
		MCWB	33.6	33.8	45.6	56.8	61.6	65.4	66.5	65.0	62.0	54.4	47.4	35.5		
	2%	DB	32.3	33.2	46.0	75.3	85.6	94.8	95.2	95.4	84.4	69.4	48.3	34.3		
		MCWB	31.5	32.2	40.6	55.1	61.7	64.3	66.3	65.3	58.9	51.7	43.9	33.2		
	5%	DB	29.9	31.3	40.2	69.8	82.0	91.0	91.6	91.8	80.0	64.3	44.7	33.1		
		MCWB	29.0	30.3	36.6	52.7	60.6	64.0	65.2	64.6	57.5	49.7	41.7	32.2		
	10%	DB	26.7	28.8	36.8	64.5	77.1	86.4	88.2	87.9	74.9	58.6	40.9	30.5		
		MCWB	25.8	27.7	34.6	50.6	58.5	62.8	64.7	63.4	56.2	47.5	38.3	29.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.6	33.9	46.7	59.3	66.3	69.4	70.7	70.1	64.7	57.3	49.9	35.7		
		MCDB	34.1	34.7	55.7	75.0	81.5	88.3	89.1	91.5	85.6	70.2	52.0	36.5		
	2%	WB	31.4	32.1	40.8	56.6	63.9	67.3	69.1	68.0	60.9	53.7	45.0	33.6		
		MCDB	32.1	32.9	45.3	71.7	80.1	86.1	86.6	87.4	77.7	65.7	47.2	34.1		
	5%	WB	28.7	30.1	36.9	54.0	61.8	65.9	67.8	66.3	59.0	51.1	41.6	31.9		
		MCDB	29.4	30.9	39.5	67.3	77.4	84.1	84.8	85.2	75.5	60.8	44.0	32.9		
	10%	WB	25.7	27.8	34.5	51.3	59.9	64.4	66.4	64.7	57.1	48.6	38.4	29.4		
		MCDB	26.6	28.9	36.3	63.2	74.5	81.9	83.1	83.1	72.0	57.4	40.5	30.3		
(35) Mean Daily Temperature Range	MDBR		15.0	16.7	16.5	21.5	24.7	25.3	24.3	26.1	25.1	20.5	13.0	13.7		
	5% DB	MCDBR	11.0	11.7	17.3	30.5	32.0	30.9	29.5	31.7	33.3	31.2	17.2	11.3		
		MCWBR	11.0	11.0	12.4	14.7	13.9	11.8	11.1	12.8	15.7	17.7	13.0	10.4		
	5% WB	MCDBR	11.4	11.1	15.7	27.6	29.0	27.0	25.4	28.6	28.6	27.0	14.8	10.4		
		MCWBR	11.4	10.8	11.8	14.2	13.5	11.0	10.1	11.9	14.5	16.3	12.2	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.326	0.384	0.427	0.412	0.410	0.412	0.426	0.373	0.355	0.334	0.302		
	taud		2.088	2.024	1.971	2.074	2.212	2.283	2.288	2.220	2.324	2.296	2.195	2.097		
	Ebn at Noon		222	247	252	257	268	269	267	255	258	239	208	199		
	Edh at Noon		26	36	46	47	43	40	40	40	33	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		442	819	1209	1507	1906	2111	2034	1742	1266	703	326	285		
	RadStd		51	59	151	148	175	154	103	125	98	100	45	57		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.58	-0.70	N/A	N/A	N/A	+189	+151	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+1.58	-0.70	N/A	N/A	N/A	+189	+151	(47)

Nomenclature: See separate page